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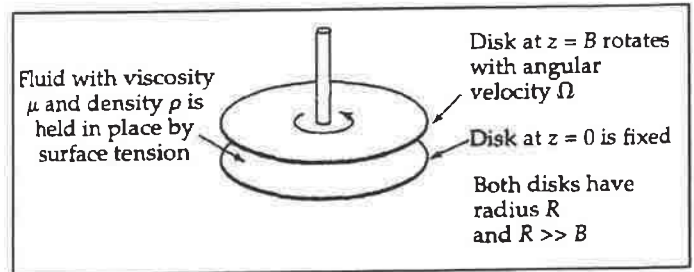
TRANSPORT PHENOMENA I, CHE 311

MIDTERM EXAM 1

CLOSED BOOK, TIME: 75 MINUTES

PARALLEL-DISK VISCOMETER

A fluid is placed in the gap between two horizontal circular disks of radius R . Disks are separated by a small distance B . The lower disk is held fixed. The upper disk is rotating at a constant angular velocity Ω .



Postulates:

- Newtonian fluid with constant density ρ and constant viscosity μ .
- $v_r = 0$ and $v_z = 0$.
- Pressure in the fluid is not a function of θ . $P(r, z)$
- Creeping flow assumption: The flow is very slow and the acceleration term $\frac{D(v)}{Dt}$ in the equation of motion is negligible.

20 1. Using the continuity equation, prove v_θ is not a function of θ at steady-state conditions.

20 2. Using the proper equation of motion at steady-state conditions, write down and simplify a partial differential equation for v_θ as a function of r and z .

20 3. Assuming v_θ may be written as r multiplied by a function of z ,
$$v_\theta(r, z) = r f(z)$$

, convert the partial differential equation for v_θ to an ordinary differential equation for f .

20 4. Knowing $v_\theta(r, z = 0) = 0$ and $v_\theta(r, z = B) = r \Omega$, write down two boundary conditions for f , solve the ordinary differential equation derived in part 1.3. and evaluate the integration constants.

20 5. One can estimate the viscosity of the fluid by measuring the torque T_z required to rotate the upper disk at a constant angular velocity. Using the velocity profile derived in part 1.4., find an equation for the viscosity in terms of T_z , Ω , B and R . Hint: torque exerted by the liquid on unit area of the upper disk is the product of the radius r (lever arm) and viscous stress in θ -direction.

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$$1. \quad \frac{\partial \rho}{\partial t} + \frac{1}{r} \frac{\partial}{\partial r} (\rho r v_r) + \frac{1}{r} \frac{\partial}{\partial \theta} (\rho v_\theta) + \frac{\partial}{\partial z} (\rho v_z) = 0$$

constant density $\quad v_r = 0 \quad \quad v_z = 0$

$$\frac{1}{r} \frac{\partial}{\partial \theta} (\rho v_\theta) = 0$$

$$\frac{\partial v_\theta}{\partial \theta} = 0$$

$$v_\theta = C + f(r, z)$$

v_θ is not dependent upon θ

$$2. \quad \rho \left(\frac{\partial v_r}{\partial t} + v_r \frac{\partial v_r}{\partial r} + \frac{v_\theta}{r} \frac{\partial v_r}{\partial \theta} + v_z \frac{\partial v_r}{\partial z} + \frac{v_r v_\theta}{r} \right) = -\frac{1}{r} \frac{\partial p}{\partial r} + \mu \left[\frac{\partial}{\partial r} \left(\frac{1}{r} \frac{\partial}{\partial r} (r v_\theta) \right) + \frac{1}{r^2} \frac{\partial^2 v_\theta}{\partial \theta^2} + \frac{\partial^2 v_\theta}{\partial z^2} + \frac{2}{r} \frac{\partial v_r}{\partial z} \right]$$

Steady State

$$0 = \mu \left[\frac{\partial}{\partial r} \left(\frac{1}{r} \frac{\partial}{\partial r} (r v_\theta) \right) + \frac{\partial^2 v_\theta}{\partial z^2} \right]$$

$$0 = \frac{\partial}{\partial r} \left(\frac{1}{r} \frac{\partial}{\partial r} (r v_\theta) \right) + \frac{\partial^2 v_\theta}{\partial z^2}$$

boundary conditions

$$v_\theta(r, z=0) = 0$$

$$v_\theta(r, z=B) = r\Omega$$

$$3. \quad v_\theta(r, z) = r f(z)$$

4.

boundary conditions for $f(z)$

$$f(z=0) = 0$$

$$f(z=B) = \frac{r\Omega}{r} = \Omega$$

$$f(z) = \frac{\Omega}{B} z$$

$$0 = \frac{d}{dr} \left(\frac{1}{r} \frac{d}{dr} (r(rf)) \right) + r \frac{d^2 f}{dz^2}$$

$$0 = f \frac{d}{dr} \left(\frac{1}{r} \frac{d}{dr} (r^2) \right) + r \frac{d^2 f}{dz^2}$$

$$0 = f \frac{d}{dr} \left(\frac{1}{r} \cdot 2r \right) + r \frac{d^2 f}{dz^2}$$

$$0 = 0 + r \frac{d^2 f}{dz^2}$$

$$\boxed{\frac{d^2 f}{dz^2} = 0}$$

$$\frac{df}{dz} = C_1$$

$$f(z) = C_1 z + C_2$$

$$f(0) = C_1(0) + C_2 = 0$$

$$C_2 = 0$$

$$f(B) = C_1 B = \Omega$$

$$C_1 = \frac{\Omega}{B}$$

$$\boxed{v_\theta(r, z) = \frac{r\Omega z}{B}}$$